3. Find the value of

(i) $\tan\left(\tan^{-1}\left(\frac{7\pi}{4}\right)\right)$ (ii) $\tan\left(\tan^{-1}(1947)\right)$ (iii) $\tan\left(\tan^{-1}(-0.2021)\right)$.

4. Find the value of (i) $\tan\left(\cos^{-1}\left(\frac{1}{2}\right)-\sin^{-1}\left(-\frac{1}{2}\right)\right)$ (ii) $\sin\left(\tan^{-1}\left(\frac{1}{2}\right)-\cos^{-1}\left(\frac{4}{5}\right)\right)$.

(iii) $\cos\left(\sin^{-1}\left(\frac{4}{5}\right)-\tan^{-1}\left(\frac{3}{4}\right)\right)$.

4.6 The Cosecant Function and the Inverse Cosecant Function

Like sine function, the cosecant function is an odd function and has period 2π. The values of cosecant function $y=\operatorname{cosec} x$ repeat after an interval of length 2π .Observe that $y=\operatorname{cosec} x=\frac{1}{\sin x}$ is not defined when $\sin x=0$. So, the **domain** of cosecant function is $\mathbb{R}\setminus\{n\pi : n\in\mathbb{Z}\}$. Since $-1\le \sin x\le 1$, $y=\operatorname{cosec} x$ does not take any value in between -1 and 1. Thus, the **range** of cosecant function is $(-\infty,1]\cup[1,\infty)$.

4.6.1 Graph of the cosecant function

In the interval $(0, 2\pi)$, the cosecant function is continuous everywhere except at the point $x=\pi$. It has neither maximum nor minimum. Roughly speaking, the value of $y=\operatorname{cosec} x$ falls from ∞ to 1 for $x\in\left(0,\frac{\pi}{2}\right]$, it raises from 1 to ∞ for $x\in\left[\frac{\pi}{2},\pi\right)$. Again, it raises from $-\infty$ to -1 for $x\in\left(\pi,\frac{3\pi}{2}\right]$ and falls from -1 to $-\infty$ for $x\in\left[\frac{3\pi}{2},2\pi\right)$.

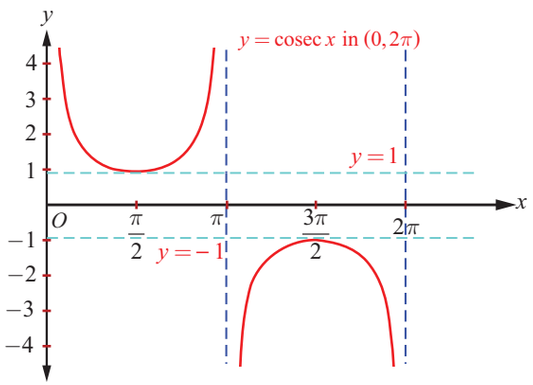


Fig. 4.19

The graph of $y=\operatorname{cosec} x,\quad x\in(0,2\pi)\setminus\{\pi\}$ is shown in the Fig. 4.19. This portion of the graph is repeated for the intervals $\cdots,(-4\pi,-2\pi)\setminus\{-3\pi\}$, $(-2\pi,0)\setminus\{-\pi\},(2\pi,4\pi)\setminus\{3\pi\},(4\pi,6\pi)\setminus\{5\pi\},\cdots$.

The entire graph of $y=\operatorname{cosec} x$ is shown in Fig. 4.20.

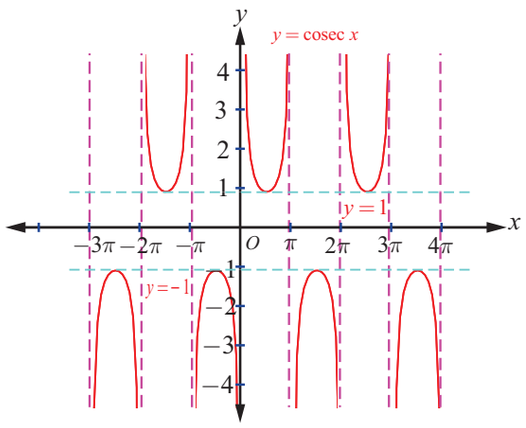


Fig. 4.20

Chapter 4 Inverse Trigonometry.indd 148 31-01-2020 17:45:39